

Project Plan

Project Innovate 2025-2026

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Introduction

This Project Plan outlines how our team intends to organise, develop, and complete the proposed campus **Emotion-Recognition Robot Project** for the Innovate 2025–2026 course.

The goal of the project is to create an interactive robot capable of **moving safely around campus, detecting a user's facial expression** once triggered, and **responding in a simple and engaging way** through speakers.

The project combines areas such as **robotics, programming, teamwork**, and **problem-solving** into a single practical system.

Here we will be describing how the team will **divide responsibilities, manage development stages, organise tasks, monitor progress**, and **reduce project risks** throughout implementations.

Therefore, this file acts as a structured guide for the development process, helping ensure that the project remains organised, achievable, and completed within the available timeframe.

Project Objectives



- **Develop a stable mobility system** – tracked-wheel robot capable of safe and reliable movements indoors.
- **Implement obstacle detection** – successful integration of the sensors to detect and avoid nearby obstacles during navigation.
- **Develop facial detection capabilities** – enable the robot to identify and capture a user's face using an integrated camera.
- **Implement basic emotion recognition** – analyse facial expressions and classify a limited set of emotions.
- **Create user interaction responses** – robot responds through audio feedback.
- **Integrate hardware and software systems** – ensure communication between sensors, controllers, and software.
- **Apply team-based development** – division of responsibilities efficiently between teams members and subgroups.
- **Deliver a functional prototype** – produce a functional enough working demonstration within the project timeframe to be able to show overall idea of the project.

Project Success Criteria



The project will be considered successful if the following outcomes are achieved:

- The robot moves safely and reliably;
- The robot can detect an object and stop/avoid it;
- Facial detection operates consistently;
- At least four emotions can be recognised correctly;
- The robot provides a clear response to the user;
- Hardware and software systems function together successfully;

Given the students lacked experience regarding this kind of project, plus the limited timeline, the project might not be concluded fully as desired but should at least show enough implementations to be able to present it at the assigned project presentation date, as an early prototype, with an appropriate explanation of what we could have further developed if time has better managed.

Project Scope



FEATURES	DESCRIPTION
TRACKED-WHEEL MOBILITY	Robot movement using a tracked chassis system
OBSTACLE DETECTION	Basic avoidance of nearby objects and walls
PUSH-BUTTON TRIGGER	User-triggered interaction mode
CAMERA-BASED FACE DETECTION	Detection and capture of user facial image for analysis
BASIC EMOTION RECOGNITION	Recognition of selected emotions (happy, neutral, sad, surprised)
AUDIO RESPONSE	Robot feedback through speakers
RASPBERRY PI AND ARDUINO INTEGRATION	Communication between processing and control systems
PROTOTYPE CONSTRUCTION	Assembly and testing of a functional robot prototype

To reduce development risks and maintain project feasibility, the team **will prioritise the completion of core functionalities before considering optional enhancements**. Additional features may only be implemented if sufficient time remains after the successful completion of the minimum viable product.

This approach helps ensure that the project **remains manageable** while still allowing opportunities for future improvements and expansion.

Team Structure & Responsibilities



The project team is structured into software, hardware, and hybrid/integration roles to **ensure efficient development, testing, and deployment** of the Emotion Reading Robot system.

- **Software team** - Focuses on programming, emotion recognition, and output response.
- **Hardware team** - Responsible for motors, sensors, wiring, and mechanical assembly.
- **Integration team** – Pi-Arduino communication, and robot control logic.

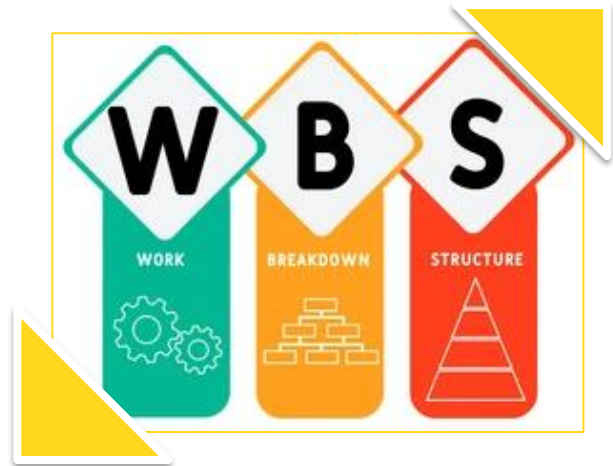
TEAM MEMBER	AREA	MAIN RESPONSIBILITY	KEY TASKS
JESSICA CAMACHO & KRISTIAN ENDREV	Software	Python related coding	Face detection Emotion recognition Text-to-speech output
HERBERT DOROTHEA & KYRA KOVACS	Hardware	Mechanical assembly	Robot chassis assembly Motor installation Driver wiring Hardware setup
FRANZ PETEV & FRANZ PETEV	Hybrid/Integration	Pi & Arduino communication	Arduino programming Obstacle avoidance Hardware validation
EVERYONE	All	Documentation Testing	Reports Documents Testing Presentation

Project Schedule & Task Planning

To support effective organisation and project monitoring, the development process has been divided into structured phases and task groups.

The following tools have been developed to **assist with scheduling, task management, and dependency tracking** throughout the project lifecycle:

- Work Breakdown Structure (WBS)



- Gantt Chart

- Network Diagram



Due to formatting and readability limitations within the main document, the **complete high-resolution planning diagrams** are provided through the team's shared Microsoft Teams project folder, reached through the links provided above.

Risks & Mitigations

Since this project combines **robotics, electronics, computer vision**, and **software integration**, several technical and practical risks must be considered during development, because identifying them will allow the team to **apply preventive measures, reduce delays** and **improve overall reliability** of the final system.

Main Project Risks

RISK CATEGORY	DESCRIPTION	POTENTIAL IMPACT	RECOMMENDED MITIGATION
SYSTEM INTEGRATION	Hardware and software components may fail to communicate correctly	Delays Unstable behaviour	Develop and test each subsystem separately before full integration
CAMERA PERFORMANCE	Poor lighting, incorrect angle, or low image quality may reduce face detection accuracy	Failed emotion analysis	Use adequate lighting, correct camera placement on mast and functional tilt functionality
EMOTION RECOGNITION ACCURACY	Facial expressions may be misclassified by the selected model	Incorrect responses to users	Begin with the recommended small set of clear emotions and improve through testing
POWER MANAGEMENT	Batteries may discharge quickly or provide unstable voltage	Unexpected shutdowns or resets	Use separate power supplies for motors and logic systems
MOVEMENT STABILITY	Uneven surfaces or rapid turning may affect balance	Robot tipping or poor control	Maintain low centre of gravity and test movement speeds
OBSTACLE DETECTION	Sensors may fail to detect nearby objects consistently	Collisions or unsafe movement	Use multiple sensors and adjust detection ranges
TIME CONSTRAINTS	Project scope may exceed available development time	Incomplete final product	Prioritize core features and optional enhancements later
MECHANICAL FAILURE	Loose wiring or weak mounts may fail during operation	Reduced reliability	Secure components properly and inspect regularly

Technical Considerations

CONSIDERATION	IMPORTANCE
SIMPLICITY OF DESIGN	Reduces build time and lowers chance of failure
MODULAR CONSTRUCTION	Easier maintenance and replacement of parts
USER SAFETY	Prevents collisions and protects users during interaction
EASE OF TESTING	Supports faster debugging and progress checks
EXPANDABILITY	Allows future features such as voice control or extra emotions
BATTERY ACCESSIBILITY	Simplifies charging and maintenance

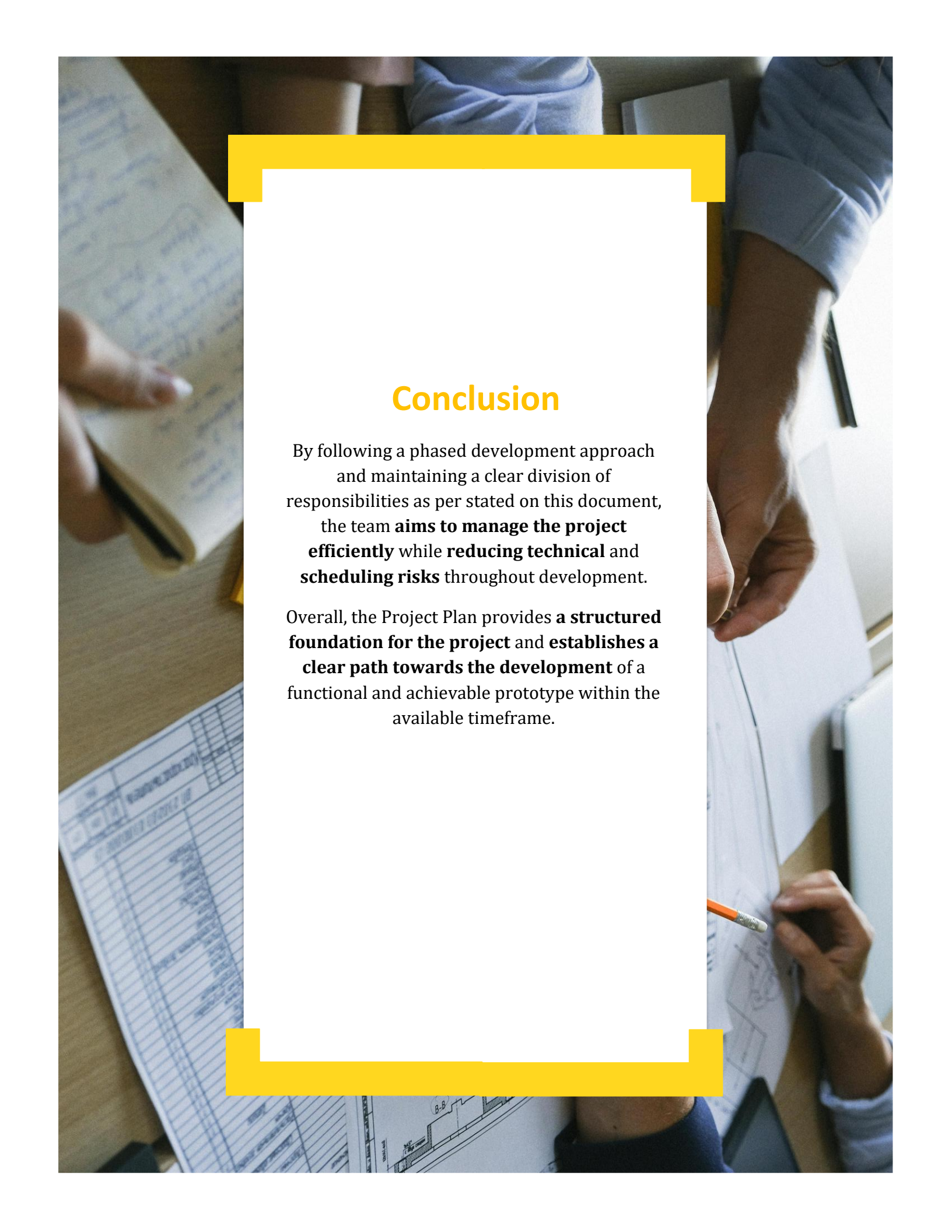
Project Scope Management

One of the most significant risks for student projects is **attempting to implement too many advanced features at the same time**. While functions such as voice commands, advanced navigation, or expanded artificial intelligence may be desirable, these **should only be pursued after** the core system is fully operational.

Overall Risk Assessment

The **overall risk level is considered manageable** if development follows the phased implementation strategy outlined in this report.

By **maintaining realistic scope, testing systems individually**, and **prioritising reliability** over unnecessary complexity, the team can significantly improve the likelihood of successful project delivery.



Conclusion

By following a phased development approach and maintaining a clear division of responsibilities as per stated on this document, the team **aims to manage the project efficiently** while **reducing technical** and **scheduling risks** throughout development.

Overall, the Project Plan provides **a structured foundation for the project** and **establishes a clear path towards the development** of a functional and achievable prototype within the available timeframe.